```
import sqlite3
import pandas as pd

# Create database connection
conn = sqlite3.connect('sales_data.db')

# Create cursor
cursor = conn.cursor()
```

```
cursor.execute("""
CREATE TABLE online_sales (
    order_id INTEGER,
    order_date DATE,
    amount REAL,
    product_id INTEGER
)
""")

conn.commit()
```

```
cursor.executemany("""
INSERT INTO online_sales (order_id, order_date, amount, product_id)
VALUES (?, ?, ?, ?)
""", [
    (1, '2024-01-05', 500, 101),
    (2, '2024-01-10', 700, 102),
    (3, '2024-02-15', 300, 103),
    (4, '2024-02-20', 450, 104),
    (5, '2024-03-08', 800, 105),
    (6, '2024-03-15', 650, 106),
    (7, '2024-04-10', 900, 107),
    (8, '2024-04-22', 400, 108),
    (9, '2024-05-05', 1000, 109),
    (10, '2024-05-18', 750, 110)
])

conn.commit()
```

```
df = pd.read_sql_query("SELECT * FROM online_sales", conn)
print(df)
```

```
   order_id  order_date  amount  product_id
0         1  2024-01-05   500.0         101
1         2  2024-01-10   700.0         102
2         3  2024-02-15   300.0         103
3         4  2024-02-20   450.0         104
4         5  2024-03-08   800.0         105
5         6  2024-03-15   650.0         106
6         7  2024-04-10   900.0         107
7         8  2024-04-22   400.0         108
8         9  2024-05-05  1000.0         109
9        10  2024-05-18   750.0         110
```

```
query = """
SELECT
    strftime('%Y', order_date) AS Year,
    strftime('%m', order_date) AS Month,
```

```
    SUM(amount) AS Total_Revenue,
    COUNT(DISTINCT order_id) AS Total_Orders
FROM online_sales
GROUP BY Year, Month
ORDER BY Year, Month
"""

sales_analysis = pd.read_sql_query(query, conn)
print(sales_analysis)
```

```
   Year Month  Total_Revenue  Total_Orders
0  2024    01         1200.0             2
1  2024    02          750.0             2
2  2024    03         1450.0             2
3  2024    04         1300.0             2
4  2024    05         1750.0             2
```

```
top_months_query = """
SELECT
    strftime('%Y', order_date) AS Year,
    strftime('%m', order_date) AS Month,
    SUM(amount) AS Total_Revenue
FROM online_sales
GROUP BY Year, Month
ORDER BY Total_Revenue DESC
LIMIT 3
"""

top_months = pd.read_sql_query(top_months_query, conn)
print(top_months)
```

```
   Year Month  Total_Revenue
0  2024    05         1750.0
1  2024    03         1450.0
2  2024    04         1300.0
```


```
top_months_query = """
SELECT
    strftime('%Y', order_date) AS Year,
    strftime('%m', order_date) AS Month,
    SUM(amount) AS Total_Revenue
FROM online_sales
GROUP BY Year, Month
ORDER BY Total_Revenue DESC
LIMIT 3
"""

top_months = pd.read_sql_query(top_months_query, conn)
print(top_months)
```

```
   Year Month  Total_Revenue
0  2024    05         1750.0
1  2024    03         1450.0
2  2024    04         1300.0
```


```
sales_analysis.to_csv("sales_analysis.csv", index=False)
top_months.to_csv("top_sales_months.csv", index=False)

print("CSV files saved successfully")
```

```
CSV files saved successfully
```

```
from google.colab import files

files.download('sales_analysis.csv')
files.download('top_sales_months.csv')
```